Introduction

The Anderson-Darling (AD) test is a goodness-of-fit test that weights the tails of the distribution more heavily than the Kolmogorov-Smirnov test. It is widely used for normality testing in engineering and quality-control contexts, where tail behaviour often determines pass/fail decisions.

Prerequisites

Empirical CDF, goodness-of-fit testing.

Theory

For a hypothesised CDF F and empirical CDF $\hat{F}_n$, the AD statistic is

$$A^2 = n \int \frac{(\hat{F}_n(x) - F(x))^2}{F(x)(1 - F(x))} dF(x).$$

The denominator $F(1-F)$ gives less weight to the middle of the distribution and more to the tails, where departures from the hypothesised distribution are often clinically or industrially important.

A computational form for a sample sorted as $y_{(1)} \leq \ldots \leq y_{(n)}$:

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^{n} (2i-1) \left[\log F(y_{(i)}) + \log(1 - F(y_{(n-i+1)})) \right].$$

Critical values depend on whether the distribution's parameters are known or estimated from the sample.

Assumptions

- iid observations.
- The hypothesised distribution is fully specified or parameters are consistently estimated.

R Implementation

```r
library(nortest)
set.seed(2026)

# Normal sample: AD should not reject
x_norm <- rnorm(100)
ad.test(x_norm)

# Heavy-tailed sample: AD should reject
x_t3 <- rt(100, df = 3)
ad.test(x_t3)

# Compare to Shapiro-Wilk
shapiro.test(x_norm)$p.value
shapiro.test(x_t3)$p.value
```

Output & Results

```
        Anderson-Darling normality test
data:  x_norm
A = 0.35, p-value = 0.47          # normality not rejected
```

```
        Anderson-Darling normality test
data:  x_t3
A = 2.10, p-value = 2.8e-05        # strongly rejects normality

Shapiro-Wilk p-values:
  normal sample: 0.42
  t(3) sample:   0.0001
```

Both tests correctly reject the heavy-tailed sample; AD has moderately higher power than Shapiro-Wilk in the tails.

Interpretation

"The Anderson-Darling test did not reject normality for the biomarker residuals ($A = 0.35$, $p = 0.47$), supporting the use of parametric methods."

Practical Tips

- AD is more sensitive to tail departures than Kolmogorov-Smirnov and often more powerful than Shapiro-Wilk for detecting heavy tails.
- `nortest::ad.test()` handles normality; `goftest::ad.test()` supports arbitrary reference distributions.
- For testing fit to a parametric distribution with estimated parameters, use the correct tabled critical values (nortest handles this for normality).
- Very large samples will reject any hypothesised distribution because real data have atoms and rounding.
- For quality-control contexts where tail behaviour governs defect rates, AD is the preferred GoF test.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests